

Hotel Reservation System

A Web-Based Hotel Room Reservation and Management System

Technologies: Java JEE / Jakarta EE · JSP/JSF · JPA/Hibernate · MySQL



What We're Building



The System

A Java JEE/Jakarta EE web application enabling customers to search, book, and manage hotel room reservations online. An integrated Admin Dashboard allows hotel staff to manage rooms, customers, and reservations.

The Problem We're Solving

- Difficulty checking room availability
- Risk of duplicate or overlapping reservations
- Time-consuming reservation management
- Difficulty tracking current and previous bookings

❏ **Solution:** A centralized hotel reservation and management system.

Two Distinct User Roles

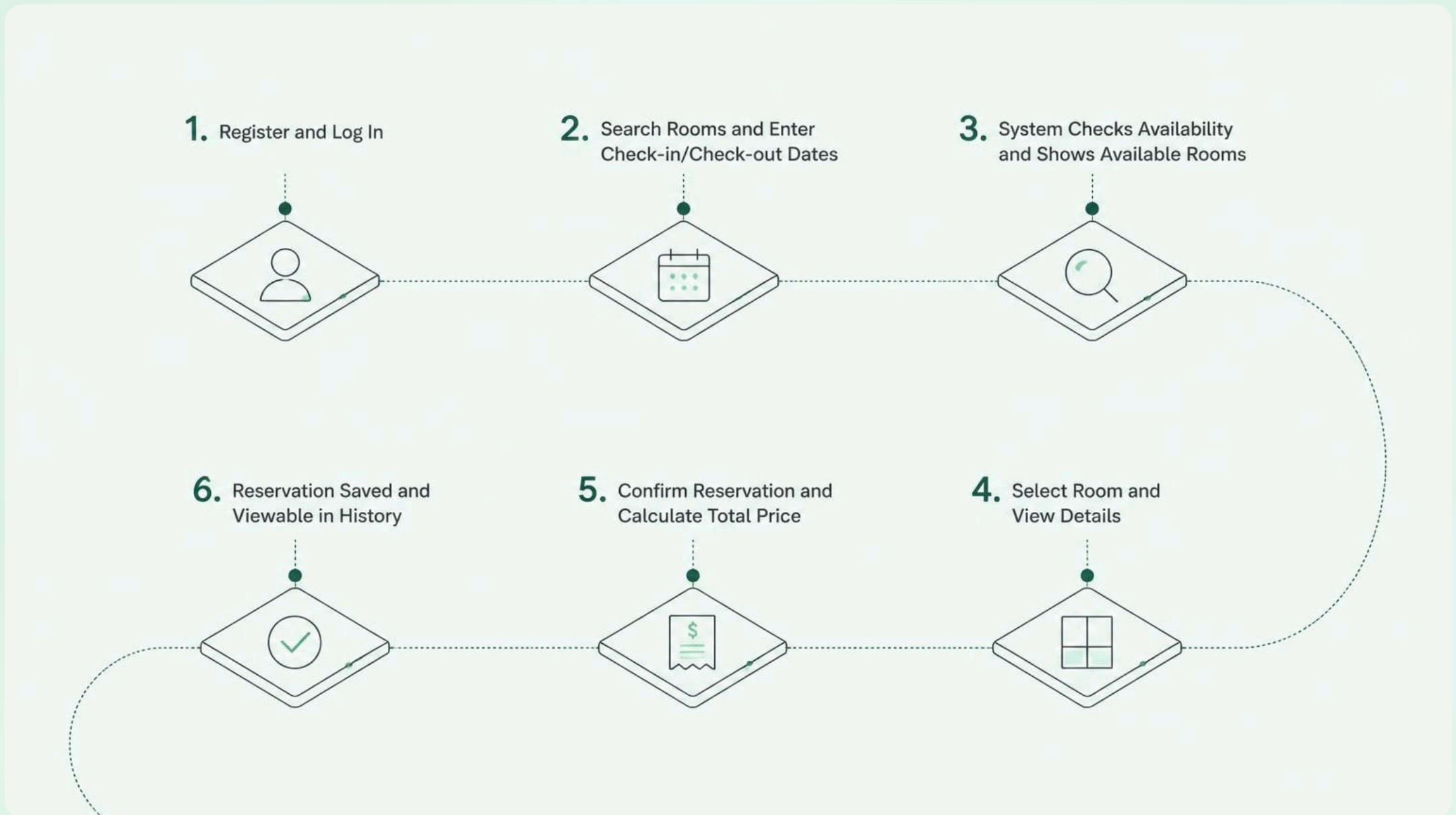
Customer

- Register and log in
- Search rooms and check availability
- View room details
- Make and confirm reservations
- View reservation history
- Cancel reservations

Administrator

- Log in to Admin Dashboard
- Manage rooms and room types
- Manage customer accounts
- Manage all reservations
- Confirm or cancel reservations
- Monitor room availability

Reservation Process



Customers complete the full reservation process in minutes. The system automatically checks date conflicts and calculates the total price before confirming the booking.

Room Availability & Price Calculation

Availability Check

The system queries existing reservations to detect date overlaps.

⚠ **Room 201:** Booked Aug 25–28. New request: Aug 26–27 → **Not Available** (dates overlap).

✅ **Room 305:** No conflicting reservations → **Available** for booking.

Rule: A room cannot be reserved by two customers for overlapping dates.

Price Calculation

Total Price = Room Price per Night × Number of Nights

Room Rate

Deluxe Room
\$80 / night

Duration

3 nights

Total

$\$80 \times 3 = \mathbf{\$240}$

Administrator Workflow & Dashboard



Administrators have full system access to manage hotel operations, while customers manage only their own reservations.

Dashboard Overview

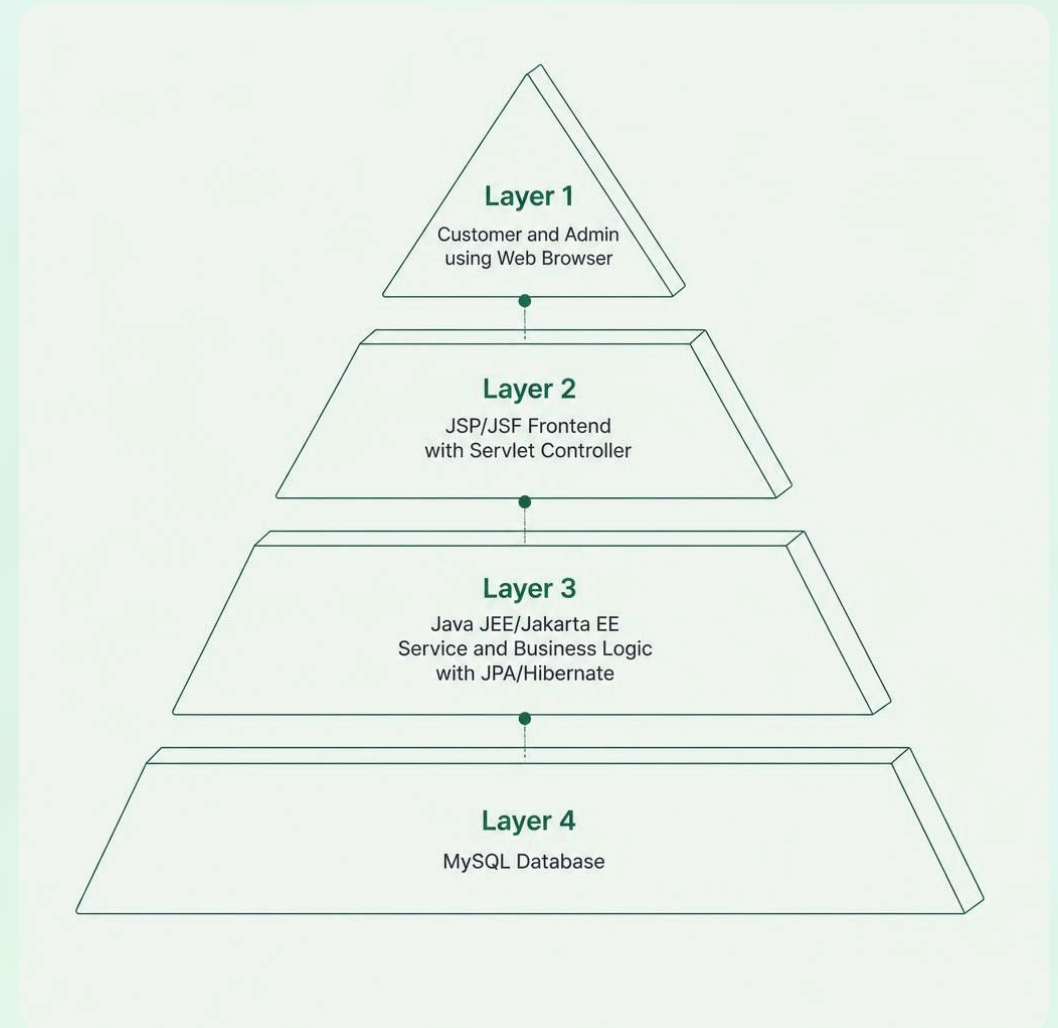


Management Sections

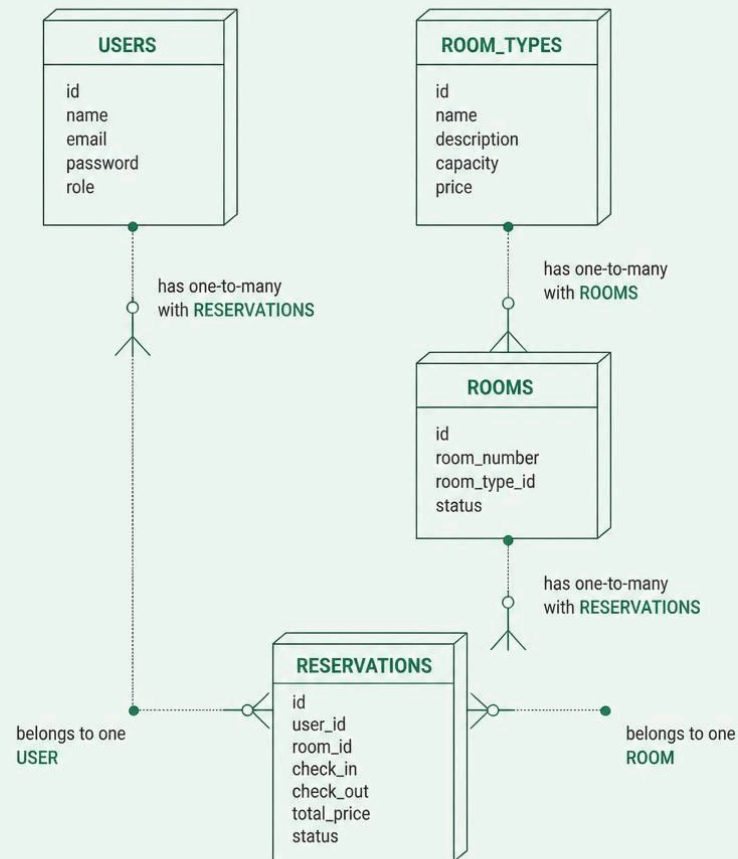
- | | |
|---------------------|------------------------|
| Room Management | Room Type Management |
| Customer Management | Reservation Management |

System Architecture

The frontend and backend are part of the same Java JEE/Jakarta EE project. The application layer handles all business logic and data persistence through JPA/Hibernate, connected to a MySQL database.



Database Structure



Entity Relationships

- **Users → Reservations:** One user can have many reservations
- **Room Types → Rooms:** One type can contain many rooms
- **Rooms → Reservations:** One room can have many reservations over different dates
- **Reservations:** Each belongs to one user and one room

Technologies Used



Frontend

HTML, CSS, JavaScript, JSP/JSF,
Bootstrap



Backend

Java, Java JEE / Jakarta EE, Servlets



Persistence

JPA / Hibernate



Database

MySQL



Build & Server

Maven · Payara / WildFly

Expected Results & Future Improvements

✓ Version 1.0 — Core Deliverables

- Customers can search and reserve rooms online
- System prevents overlapping reservations
- Customers can manage their own reservations
- Admins can manage rooms, customers, and bookings
- All data stored in a centralized MySQL database

Future Improvements

- Online payment processing
- Email reservation confirmation
- PDF booking receipts
- Room images and customer reviews
- Discount codes and advanced reports
- Mobile application

Future features are out of scope for the initial version.